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ML2MAS: a multi-agent reinforcement learning and BNNs-GAN integration framework for smart manufacturing optimization

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ABSTRACT

The rise of smart manufacturing, driven by advanced technological enablers, has transformed traditional production processes, creating new opportunities for operational efficiency and predictive maintenance (PdM). This study introduces ML2MAS, an innovative framework that integrates machine learning (ML) models within a multi-agent reinforcement learning (MAS-RL) system to enhance PdM capabilities in manufacturing environments. ML2MAS combines Bayesian Neural Networks (BNNs) for handling high-dimensional data and quantifying prediction uncertainty alongside Generative Adversarial Networks (GANs) for synthetic data generation, addressing the challenge of limited labelled datasets and improving model robustness. Integrating these components within MAS enables PdM to make decentralized, real-time decisions. Empirical results from a case study demonstrate substantial improvements, achieving a 99 % F1-score in predictive accuracy and notable reductions in maintenance costs. The proposed ML2MAS framework ensures a cohesive, adaptive PdM solution and contributes to more sustainable and efficient manufacturing operations.

1. Introduction

Industry 4.0 has revolutionized manufacturing by introducing smart technologies that enhance operational efficiency and data-driven decision-making [1,2]. One of the critical functions within this paradigm is predictive maintenance (PdM), which focuses on anticipating machine failures before they occur, thereby minimizing costly downtime and optimizing the utilization of resources. However, traditional PdM strategies—such as routine inspections or threshold-based alerts—are often reactive, rigid and unable to accommodate the complexity and dynamic nature of modern smart manufacturing environments [3–6].

While recent efforts have introduced AI and ML techniques into PdM to enhance fault prediction accuracy, several key limitations remain. First, the performance of ML models is highly dependent on the availability of large, high-quality labelled datasets. In many industrial applications, labelled failure data are scarce due to the rarity of critical faults and the high cost of manual annotation. Second, many existing frameworks are based on centralized decision-making models, which are not well suited for the distributed, modular nature of smart manufacturing systems (SMSs). Finally, many PdM models are not robust against non-stationary data streams—that is, they lack the adaptability to respond to evolving equipment behaviour and changes in operating conditions.

To address these challenges, we propose ML2MAS, a novel multi-agent reinforcement learning framework for PdM in smart manufacturing. ML2MAS integrates several advanced AI technologies within a decentralized architecture:

- Bayesian Neural Networks (BNNs) are employed to quantify uncertainty in predictions, improving reliability and reducing overfitting [7].
- Generative Adversarial Networks (GANs) are used to augment the training dataset by generating high-fidelity synthetic sensor data that reflects real-world operating conditions [8].
- A Multi-Agent System (MAS) structure enables agents to collaborate and make localized decisions in real time, improving system responsiveness and scalability [9].

This agent-based architecture aligns well with modern manufacturing infrastructures composed of cyber-physical systems (CPS), Industrial IoT devices and modular automation platforms [10,11]. Each agent is responsible for distinct tasks such as data preprocessing, fault prediction, model training and maintenance decision-making.

To validate the effectiveness of ML2MAS, we conduct a detailed case study using a publicly available dataset of hydraulic system sensor data. The experimental results show that ML2MAS significantly improves PdM performance metrics such as predictive accuracy and F1-score, while reducing downtime and enhancing overall system efficiency [12,13]. The integration of uncertainty-aware learning and synthetic

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data generation allows the system to generalize better in data-scarce environments.

ML2MAS offers a unified, adaptable and scalable framework that addresses the core limitations of existing PdM approaches. Its key contributions include:

- A hybrid learning pipeline that combines real and synthetic data using GANs to overcome data scarcity
- An uncertainty-aware decision-making mechanism via BNNs to improve prediction confidence
- A decentralized, cooperative agent system that supports dynamic adaptation to real-time data and operational changes

This integrated approach supports proactive, sustainable maintenance strategies and aligns with broader goals such as minimizing resource waste and advancing the UN Sustainable Development Goals (SDGs) in industrial systems [14,15].

2. Literature review

Recent advances in smart manufacturing have accelerated the integration of intelligent technologies such as ML, MAS and edge computing into industrial processes. As smart factories generate vast amounts of real-time sensor data, there is an increasing need for efficient, scalable and adaptive PdM frameworks. This literature review explores current research related to the key components of our proposed framework—namely, ML models, GANs and multi-agent reinforcement learning systems—and identifies gaps that ML2MAS seeks to address.

2.1. Manufacturing systems

Manufacturing, at its core, is transforming raw resources into finished goods, a broad and nuanced process. Classical mechanical manufacturing, often termed “special manufacture”, involves intricate processing and assembly processes. However, the scope of manufacturing extends far beyond these processes, encompassing a life cycle that includes product development, material selection, production scheduling, execution and after-market services such as sales and customer support [16].

Modern manufacturing systems are evolving toward intelligent, adaptive architectures enabled by Industry 4.0 technologies. These include AI-powered systems that improve responsiveness, decentralization and scalability—laying the foundation for advanced PdM frameworks [17,18].

2.2. ML models in manufacturing

In factories, enterprise resource planning (ERP) systems manage data across various departments, such as accounting, production, supply chain and human resources. While these systems, such as those provided by SAP, may require significant storage capacity (e.g., a minimum of 60 GB), the data generated from IoT-enabled devices like sensors, solenoids and actuators can be even more substantial, often producing around 5 GB of data weekly. Despite advanced clustered processors, extracting, transforming and loading (ETL) this structured and unstructured data into a data warehouse can be time-consuming, leading to what is known as ETL lag.

To mitigate ETL lag and enable near real-time ML at manufacturing sites, a lambda (λ) architecture is often employed. This architecture allows quick preliminary insights from partial data and more accurate analyses from complete datasets. Data management within this system involves breaking down large volumes into manageable chunks using systems like Kafka for queuing and Storm, Spark or Flink for streaming [19]. Data flow through this architecture, refined by algorithms at various stages, continuously retrains and adjusts DL model parameters for improved predictive accuracy.

The lambda architecture is particularly beneficial for PdM. It allows manufacturers to analyze incoming sensor data in real-time, identifying patterns that might indicate potential failures. This approach supports proactive maintenance by using ML models to detect anomalies and predict equipment issues before they result in costly downtime [20]. By facilitating both speed and batch layers of analysis, the architecture ensures a balanced trade-off between processing speed and accuracy.

The most accurate ML algorithms operate within batch layers, which handle the complete dataset. However, the precision of these batch layers comes with high data storage and computation costs. A speed layer processes only the most recent data to enhance operational efficiency, facilitating faster computations and data uploads. Systems like Cassandra or HBase, used in these layers, support near RT control by incrementally updating views generated by analytics programs.

With the evolution of cloud technologies, smart manufacturing increasingly relies on cloud-based infrastructures [21]. Significant investments are channelled into Infrastructure as a Service (IaaS) and Software as a Service (SaaS), with projections showing a substantial shift toward cloud storage by 2025 [21–23]. Major cloud providers now offer automated ML tools, making AI more accessible to various organisations [24]. These cloud-based systems can support PdM by providing the computational power needed for complex model training and big data analysis, even for small and medium-sized enterprises.

Despite the advantages of scalable cloud resources and faster network speeds, such as those provided by 5 G, latency concerns remain, especially when transmitting data across extensive networks [25–27]. To address these challenges, edge computing is increasingly used to move data storage and processing closer to the source of data generation. This approach alleviates latency issues and supports the real-time processing needs of CPS in manufacturing environments, where local responsiveness to ML-driven data insights is crucial.

In the context of predictive maintenance, edge computing allows for real-time anomaly detection and decision-making, ensuring that potential equipment failures are addressed promptly [28]. For instance, using deep neural networks (DNNs) at the edge enables manufacturers to analyze sensor data streams directly from machinery, identifying faults before they propagate. This is particularly beneficial for time-sensitive applications, such as monitoring critical equipment like hydraulic pumps or robotic arms.

In addition to traditional ML models like Support Vector Machines (SVMs) and Random Forest (RF), DL models have proven particularly effective in handling high-dimensional data from manufacturing processes. Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) can be employed for time-series data analysis, which is common in sensor data streams. At the same time, GANs can be used for data augmentation to improve model robustness when labelled data is limited [8]. These models enhance predictive capabilities, allowing manufacturers to detect early signs of potential failures and optimize maintenance schedules.

A practical example is the integration of GANs with real-time data processing at the edge, which enables the generation of synthetic sensor data to train predictive models under data-scarce conditions. This combination of cloud, edge and advanced ML models provides a comprehensive framework for enhancing PdM, ultimately reducing equipment downtime and maintenance costs while increasing overall operational efficiency.

2.3. Multi-agent systems reinforcement learning (MAS-RL) in manufacturing

Multi-agent systems reinforced with RL (MAS-RL) enable multiple intelligent agents to collaborate in optimizing system performance, especially in complex and dynamic manufacturing environments [29]. In this study, we refer to MAS-RL as the core of our proposed ML2MAS framework, where agents with distinct roles—such as monitoring, pre-

diction and decision-making—learn to cooperate through reinforcement signals.

MAS-RL is particularly relevant for PdM, as it allows decentralized agents to adapt to local machine conditions while aligning with global maintenance objectives [7]. In smart manufacturing contexts, this decentralization is critical for scalability and responsiveness. Agents learn policies through interaction with the environment and with one another, adapting in real-time to equipment behaviour, production demands and non-stationary data [18].

A detailed explanation of the RL foundations, including Markov Decision Processes (MDPs) and partially observable MDPs (POMDPs), is provided in Section 4.

The following subsection presents recent studies exploring MAS and ML integration for PdM in industrial applications.

2.4. Related work

Industry 4.0 has precipitated a paradigm shift in manufacturing practices, with a burgeoning emphasis on PdM facilitated by AI and ML [30]. PdM has emerged as a noteworthy method, particularly for its capacity to foresee equipment malfunctions before they happen. This technique entails gathering and analysing data from diverse sensors to detect patterns signalling imminent equipment failures. By implementing PdM, improvements can be achieved in production quality, system control and cost management, as well as reducing the downtime of machinery or systems [31]. This literature review integrates recent studies exploring MAS and ML applications in PdM and integrating these technologies within manufacturing environments.

2.4.1. MASs in smart manufacturing

Multi-agent systems (MAS) consist of autonomous agents that interact with each other and their environment to perform distributed tasks. In manufacturing, MAS are used for process coordination, fault detection and real-time decision-making. Benjamin Nitsche et al. [18] investigate MAS's potential in automating logistics and supply chain network decision-making. It examines 11 MAS use cases, assessing their benefits in enhancing network resilience and organisational productivity and identifies barriers such as lack of standardisation, technological maturity and high costs. In [29], the study introduced an innovative MAS for learning maintenance policies amidst uncertainties of multiple machine failures. Their approach, incorporating RL agents for dynamic maintenance task assignment, demonstrated a remarkable 75 % improvement in performance over traditional preventive and corrective maintenance policies, as evidenced by reduced downtime and enhanced failure prevention.

2.4.2. ML models for PdM

ML models are widely used in PdM to detect patterns in equipment data and predict failures. These models include supervised techniques like SVMs and ensemble methods and are often enhanced by deep learning (DL) architectures for time-series analysis [32]. A comprehensive review of AI-based models for PdM of water injection pumps in the oil and gas industry was provided in [33]. It covers the selection of algorithms, data requirements and optimisation strategies, highlighting the effectiveness of these models in enhancing pump performance and predicting maintenance needs. The review emphasises the importance of data quality and interpretability in AI integration for maintenance. It serves as a valuable resource for guiding future research and improving maintenance practices in the industry. Furthermore, the study of [34] proposes a novel method, combining digital twins and PdM with ML, to enhance robotic cell reliability. Their case study on a spot-welding robotic cell demonstrates optimised reliability through real-time detection and classification of faulty stepper motor bearings, estimating their remaining useful life (RUL).

2.4.3. Integration of ML models with MASs

Integrating ML with MAS enables intelligent agents to not only collaborate but also learn and adapt their behaviours based on changing data and performance feedback. The integration of ML models with MASs in manufacturing has been the focus of several recent studies [35]. Yang et al. [36] created a game-based collaborative method for planning integrated electricity-gas energy systems, considering wind power uncertainty. This approach balances stakeholder profits and system security, effectively managing energy uncertainties and ensuring equitable outcomes. An innovative approach by Hessen et al. [37] developed a multi-agent e-learning system to study agent interactions in educational settings. The system includes data collection, pre-processing, agent integration, feature extraction, optimisation and training of ML algorithms and performance assessment. It uses a left join for student data integration and an Extra Trees Classifier for feature selection, with results showing high performance in cross-validation and testing.

2.4.4. GANs in smart manufacturing

Generative Adversarial Networks (GANs) have gained attention in the manufacturing domain for their ability to generate synthetic data that closely resembles real sensor data [38]. GANs consist of two neural networks—a generator and a discriminator—that work in opposition, with the generator creating synthetic data and distinguishing between real and synthetic data [36,37].

In PdM, GANs are employed to address the issue of limited labelled data, which can hinder the performance of ML models. By augmenting the training dataset with synthetic data, GANs improve the robustness and accuracy of ML models, enabling them to perform better under data-scarce conditions [8,38]. This makes GANs an invaluable tool for improving predictive maintenance in smart manufacturing environments, where real-world sensor data may be limited or imbalanced [8].

Integrating GANs with MAS in predictive maintenance frameworks has shown promising results. This allows manufacturers to leverage real and synthetic data for more accurate and reliable predictions. This approach enhances the ability of ML models to detect early signs of equipment failure, thus reducing unscheduled downtime and maintenance costs [8,39].

2.4.5. Challenges and opportunities

Despite the advancements in ML/DL, MAS and GAN technologies, several challenges remain in the context of PdM in smart manufacturing [39]. One of the key challenges is the interpretability of ML models. While these models are highly effective at predicting equipment failures, their decision-making processes are often opaque, making it difficult for operators to understand why certain predictions are made [3].

Another challenge is integrating MAS with existing manufacturing systems, particularly legacy equipment. Ensuring seamless communication between agents and machines is critical for the success of MAS-driven predictive maintenance systems [34]. Additionally, the scalability of MAS frameworks in larger, more complex manufacturing environments remains an area that requires further research [18].

On the other hand, the integration of GANs for data augmentation presents significant opportunities for improving PdM. By addressing the issue of limited data, GANs can enhance the performance of ML models and make predictive maintenance systems more reliable and robust. Furthermore, the continued development of MAS and RL algorithms offers new possibilities for creating adaptive, decentralized systems that can autonomously optimize maintenance strategies in real-time [38].

Table 1 provides a comparative overview of recent PdM frameworks, highlighting key differences in architecture, decision logic and performance. This contextualizes the novelty of ML2MAS, which uniquely combines decentralized multi-agent learning, uncertainty estimation via BNNs and data augmentation through GANs.

This comparison underscores how ML2MAS addresses gaps in previous works by integrating uncertainty-aware decision-making and scal-

Table 1
Comparative analysis of recent PdM frameworks.

Framework / study	Data strategy	Architecture	Decision-making	Reported performance	Limitations
Traditional ML (e.g., SVM, RF) [7,33]	Real sensor data only	Centralized	Threshold-based or supervised ML	~85–90 % F1	Needs labelled data; lacks adaptability
GAN-enhanced CNNs [8,38]	Real + GAN synthetic data	Centralized	ML-based prediction	Up to 95 % (variable)	No real-time adaptation; ignores uncertainty
MAS with RL (e.g., DQN) [29]	Real data only	Decentralized MAS	RL (DQN, basic multi-agent)	~75 % improvement vs traditional maintenance	No uncertainty estimation; partial coordination
ML-integrated MAS [35,37]	Real data only	MAS + ML	ML-driven decision support	High performance in specific domains	Limited PdM focus; lacks data augmentation
ML2MAS (This Work)	Real + GAN synthetic data	Fully decentralized MAS-RL	RL (PPO) + BNNs (uncertainty-aware)	F1-scores: 0.95–0.99	Early stage; scaling to complex setups ongoing

able, agent-based coordination—capabilities not simultaneously offered by earlier models.

3. Conceptual framework

While the integration of ML models has garnered attention, the role of MASs in optimising this integration remains underexplored [35]. This section outlines the proposed conceptual framework for integrating ML/DL models, MAS-RL and GANs for PdM in smart manufacturing environments. The conceptual framework introduced here seeks to integrate these elements, offering a robust and adaptive approach to enhancing PdM in smart manufacturing environments.

3.1. Components of the framework

1. ML models:

- Role: ML models analyse sensor data and predict equipment failures. They handle high-dimensional data and provide accurate predictive analytics.
- Integration: BNNs and other ML models are integrated within the MAS to process data and generate maintenance insights. GANs are used for data augmentation, improving the robustness of ML models by generating synthetic training data.

2. MAS:

- Role: MAS orchestrates the different components of the PdM system, ensuring seamless interaction and real-time processing.
- Structure: The MAS comprises various specialised agents, including “Data Pre-processing Agents”, “Sensor Agents”, “Aggregator Agents”, “Predictor Agents”, “Model Training Agents”, “Decision Maker Agents” and “User Interface Agents”.
- Function: Each agent performs distinct tasks such as data collection, pre-processing, model training and decision-making, contributing to a decentralised and collaborative system.

3. RL:

- Role: RL algorithms enable the system to learn from interactions with the manufacturing environment, continuously optimising PdM strategies.
- Implementation: Agents use RL to adjust their actions based on feedback, enhancing the system’s adaptability and decision-making accuracy.

4. GANs:

- Role: GANs generate synthetic data to augment the training datasets, addressing the limitations posed by limited or imbalanced real-world data.
- Integration: Synthetic data generated by GANs enhances the ML models’ training process, improving prediction accuracy and robustness.

3.2. Interaction and workflow

1. Data flow:

- Data Pre-processing Agents (DPAs) such as Sensor Agents to collect real-time data from manufacturing equipment and Aggregator Agents to preprocess and organise the data for analysis.
- Predictor Agents use ML models to predict equipment conditions and potential failures.
- GANs generate synthetic data to augment the training datasets.
- Model Training Agents (MTAs) continuously update ML models using new data, including synthetic data from GANs and RL techniques. During training, agents prioritized real data unless GAN-generated samples offered better representation for minority fault categories. A confidence-weighted strategy ensured that synthetic data was only included when it enhanced class balance or diversity, maintaining model robustness.

2. Decision-making process:

- Decision-Maker Agents (DMAs) evaluate predictions and uncertainties to make informed maintenance decisions. BNNs generated predictive distributions instead of single outputs. DMAs interpreted the variance of these distributions as uncertainty and decisions were made only when predicted class probabilities exceeded a confidence threshold (e.g., 0.85), ensuring safe and reliable maintenance actions.
- User Interface Agents present data and insights to human operators (stockholders), facilitating real-time monitoring and adjustments.

3. Feedback loop:

- The system incorporates a feedback loop in which maintenance actions and outcomes are fed back into the ML models, GANs and RL algorithms, refining predictions and improving system performance over time. This feedback mechanism also allows the system to handle non-stationary data streams by retraining models using both recent real-time inputs and synthetic data. As a result, agents can detect and adapt to concept drift and develop operational patterns within the manufacturing environment.

Fig. 1 illustrates the architecture of the proposed ML2MAS framework, highlighting the interaction between its core components. The framework operates through multiple intelligent agents, each assigned a distinct role—ranging from data sensing and preprocessing to model training and decision-making. Real-time sensor data is fed into the system and optionally augmented by synthetic data generated through GANs to improve data diversity and address class imbalance. The MTA utilizes RL (specifically PPO) and BNNs to optimize predictive accuracy while quantifying uncertainty. These predictions are then evaluated by the DMA, which incorporates uncertainty estimates into maintenance decisions. The decentralized structure of the framework allows agents

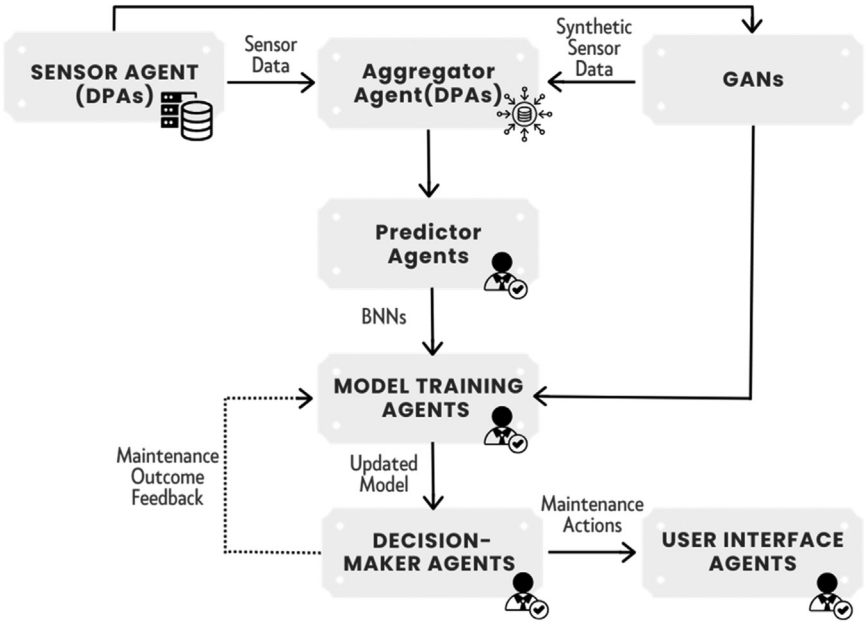


Fig. 1. The architecture and data flow of the proposed ML2MAS framework.

to collaborate asynchronously, adapt to changing conditions and continuously refine their strategies through feedback loops.

This conceptual framework sets the stage for a comprehensive and integrated approach to PdM, leveraging the latest advancements in ML, MAS-RL and GANs to optimise maintenance practices and improve manufacturing efficiency.

4. Theoretical considerations

This section provides the mathematical foundations and model selection rationale behind the ML2MAS framework. While Section 2 reviewed the applications of GANs, RL, BNNs and MAS in existing research and Section 3 introduced our system design, this section presents the formal underpinnings of the algorithms used. We focus on the theoretical principles that guide our integration of these models within the ML2MAS framework.

4.1. Reinforcement learning foundations

Reinforcement Learning (RL) enables agents to learn optimal decision-making strategies based on interactions with the environment. This section outlines the core mathematical foundations relevant to ML2MAS, including Q-learning, multi-agent stochastic games and value function updates used in our framework.

In essence, RL can be viewed as the most general form of learning problems. Contrary to supervised machine learning, the target for an RL algorithm is feedback, which is partial and almost a delayed reward or penalty. Moreover, RL differs from unsupervised learning since its primary objective is to maximise reward signals rather than discover hidden structures within unlabelled data. Based on Sutton and Barto [40], an RL system's ultimate aim is to maximise the expected value of the cumulative sum of a reward (immediate reward) signal. RL generally aims to achieve a long-term goal; therefore, the reward is back-propagated and discounted by a discount factor. The goal is to maximise the discounted cumulative future rewards at a discounted rate [41].

RL problems are usually formalised using MDPs. MDPs are the mathematical representations of RL problems that originate in dynamic systems. The Markov property implies that all relevant information for a decision is encoded in the state vector s_t. An MDP problem is described by the four-tuple M = [S; A; p; r], where S denotes the environment's state space and A denotes the agent's action space at ∈ A. In s_t, the agent

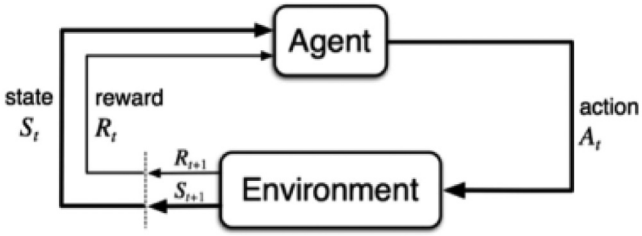


Fig. 2. Agent in loop [42].

performs to move to s_{t+1} and is immediately rewarded with r_t. Additionally, p_{ss} represents the probability that a particular action performed at state s_t leads to the state s_{t+1} being reached. The uncertainty, however, is represented by a probability distribution function that indicates whether taking the action at the state s_t results in s_{t+1} for the agent. It emphasises that the agent's actions partially influence the agent's state transitions. Fig. 2 illustrates the agent in an RL problem.

A more realistic version of MDP is the partially observable MDP (POMDP), wherein, as opposed to MDP, the agent only obtains a partial observation from the environment, leading to higher uncertainty.

The most popular RL techniques are Q-learning (QL), SARSA and actor-critic (AC), each with variants to enhance their performances in specific ways [43]. In QL, the objective is to learn an action selection policy π_t (a|s) toward maximising the reward. The policy π_t (a|s) represents the probability distribution $a_t = a$ if $s_t = s$. The Q-function Q: $S \times A \rightarrow$ R yields the discounted cumulative reward with the discount factor γ, as shown in Eq. (1), as outlined by Canese et al. [44]:

$$Q(s,a) = r + \gamma \max_{a'} \left(s', a'\right) \tag{1}$$

The QL algorithm is implemented to obtain the best action value function Q*(s, a), as shown in Eq. (2):

$$Q^*(s,a) = \max_{\pi} E\left[\sum_t r_t\, \gamma^t | s_t = s,\ a_t = a, \pi\right] \tag{2}$$

Q-values are updated using Eq. (3) every time an action is taken.

$$Q_{t+1}(s,a) = Q_t\left(s_t, a_t\right) + \alpha(r_{t+1} + \gamma \max_a Q_t\left(s_{t+1}, a\right) - Q_t\left(s_t, a_t\right) \tag{3}$$

With the emergence of DNNs, reinforcement learning has expanded to a broader range of applications by generalizing from past experiences

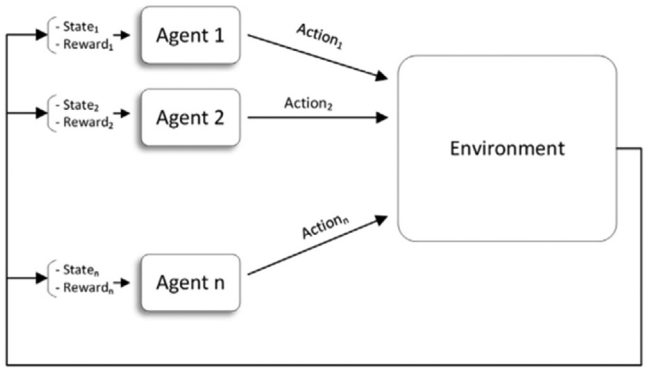


Fig. 3. Interaction between agents and environment in the MAS-RL framework [45].

to unseen scenarios. DNNs can store values and approximate RL values for unseen situations. The well-known deep Q-network (DQN) and various AC methods rely upon DNNs. DNNs are one of the enabling technologies for MAS-RL since the vast state space of MARL problems cannot be contained in the tabular form.

In MAS-RL, the actions taken by each agent change the environment and consequently change the perception of the other agents from the environment. This problem is known as non-stationarity. Fig. 3 illustrates a typical MAS-RL environment, where multiple agents interact with the environment and each other. Following the application's objectives, this interaction may be cooperative, competitive, or mixed. This research focuses primarily on cooperative MAS-RL, where the agents cooperate to maximise a standard global reward while trying to maximise their local rewards.

In the multi-agent setting, as shown in Eq. (4), the generalisation of MDP is the stochastic game. Based on Buşoniu et al. [46], a stochastic game is described via a tuple (X, U_1, …, U_n, f, ρ_1, …, ρ_n), where n denotes the number of agents; X is the environment state space; and U_i, i = 1, …, n represents the agents' action space, leading to a joint action set $U = U_1 \times \cdots \times U_n$. The state transition probability function is defined as f: $X \times U \times X \rightarrow [0, 1]$ and the rewards for the agents are ρi: $X \times U \times X \rightarrow R$, i = 1, …, n.

The state transitions in the multi-agent case result from the joint actions of the agents at step k, $u_k = [u^T_{1,k}, u^T_{n,k}]^T$, $u_k \in U$, $u_{i,k} \in U_i$.

The transpose of a vector is indicated by T. The joint policy h_i: $X \times U_i \rightarrow [0, 1]$ represents all the policies. The joint action leads to the reward $r_{i,k+1}$; thus, the returns depend on the joint policy.

$$R_i^h(x) = E\left\{\sum_{k=0}^{\infty} \gamma^k r_{i,k+1} | x_0 = x, \boldsymbol{h}\right\} \quad (4)$$

Depending on the specific application, MAS-RL is generally formalized using multi-agent (partially observable) MDPs or decentralized MDPs. These models guide how agents collaborate and optimize their strategies over time.

These foundational concepts support the learning and coordination mechanisms implemented within the ML2MAS framework, enabling agents to adapt in decentralized, dynamic environments.

4.2. Distributed artificial intelligence (DAI)

- Concept: DAI is a branch of AI that deals with systems composed of multiple interacting agents, each with capabilities and knowledge [47].
- Relevance to PdM: DAI principles are critical for developing a decentralised and collaborative PdM system. By distributing the decision-making process across multiple agents, the system can operate more efficiently and robustly, even in complex and dynamic manufacturing environments.
- Application: In our framework, DAI ensures that each agent operates autonomously while collaborating with other agents to achieve the common goal of PdM. This approach enhances system scalability and adaptability.

4.3. ML models

- Concept: As discussed in Section 2.2, ML models are essential for analyzing sensor data and predicting failures. Here, we elaborate on the theoretical role of BNNs in handling uncertainty and GANs in augmenting limited datasets [48].
- Relevance to PdM: ML models are widely used in PdM to analyze sensor data and detect anomalies. In ML2MAS, BNNs are used for their ability to quantify uncertainty in predictions, while GANs augment training datasets where real-world data is scarce or imbalanced. This combination enhances both model robustness and predictive accuracy.
- Application: We use BNNs for their ability to quantify prediction uncertainties and GANs for data augmentation to enhance training datasets. This combination helps improve the accuracy and robustness of PdM models.

4.4. RL

- Concept: As covered earlier, RL provides the foundational learning strategy for agents in ML2MAS, enabling them to continuously optimize performance based on environmental feedback [45].
- Relevance to PdM: RL allows the system to adapt its behaviour based on environmental feedback, optimising maintenance strategies over time.
- Application: RL algorithms train agents to optimise the proposed framework's predictive models and maintenance schedules. This ensures that the system remains responsive to changing conditions and can continually improve its performance.

Particularly, we used Proximal Policy Optimization (PPO) to train agent policies due to its stability in high-dimensional control tasks and superior sample efficiency in multi-agent environments. The reward function was designed to promote accurate predictions, reduce unnecessary maintenance actions and encourage high-confidence decisions. Agents were positively rewarded for correct classifications, penalized for false positives/negatives and further influenced by prediction uncertainty, discouraging decisions based on low-confidence outputs.

4.5. GANs for data augmentation

As introduced in Section 2.4.4, GANs are used in this study to address data scarcity through synthetic data augmentation. The architectural and implementation details—along with validation techniques and model training—are presented in Section 5.1.2.

4.6. MAS design and collaboration

As introduced in Section 3.1.2, the ML2MAS framework employs a layered multi-agent architecture comprising DPAs, MTAs and DMAs. The theoretical foundation for this agent-based system lies in DAI and collaborative optimization principles, which enable decentralized processing and decision-making. Each agent operates autonomously within its specialized domain while maintaining coordination through predefined communication protocols and shared objectives. This layered structure ensures scalability, adaptability and responsiveness across dynamic manufacturing environments.

4.7. Collaborative optimization

- Concept: Collaborative optimisation involves multiple agents working together to achieve a common objective, optimising their actions and interactions [49].

Table 2
Summary of the hydraulic system condition monitoring dataset.

Aspect	Details
Source	Zenodo
Application	Condition monitoring of hydraulic systems.
Data collection	Sensor data collected at 1 Hz, 10 Hz and 100 Hz frequencies. Data spans over 2205 cycles. Each cycle lasts 60 s.
Sensor measurements	1. Cooler conditions 2. Valve conditions 3. Internal pump leakage 4. Hydraulic accumulator 5. System stability
Features extracted	Statistical measures (mean, standard deviation, minimum, maximum, sum, median, skewness, kurtosis).

- Relevance to PdM: This concept is vital for ensuring that all agents in the MAS work synergistically, leveraging their collective capabilities to improve PdM outcomes.
- Application: The framework employs collaborative optimisation techniques to coordinate data processing, model training and decision-making. This leads to more effective and efficient maintenance interventions.

4.8. Transition to practical implementation

The ML2MAS framework integrates machine learning models, reinforcement learning, and GANs within a multi-agent system, with agent roles and collaboration strategies governed by principles of DAI and collaborative optimization.

The system architecture facilitates seamless data flow: real-time sensor data is first processed by DPAs, then passed to MTAs and finally interpreted by DMAs. GANs support the system by generating synthetic data for augmentation, particularly in cases of limited labelled data. A continuous feedback loop supports ongoing learning and adaptation.

This theoretical architecture supports real-time predictive maintenance that is scalable, decentralized and highly responsive. With this foundation established, the following section presents a practical case study where ML2MAS is applied to a real-world hydraulic system dataset. The implementation highlights the system's effectiveness in improving maintenance accuracy, responsiveness and operational efficiency in a smart manufacturing setting.

5. Case study: enhancing PdM in manufacturing through a multi-agent-based framework —ML2MAS

This section presents the implementation of the ML2MAS framework using a real-world hydraulic system dataset. We employed the agentMET4FOF platform [50] and utilized a publicly available dataset from Zenodo [51] focused on condition monitoring of hydraulic components. The aim of this case study is to demonstrate how the integration of ML models, GAN-based data augmentation and multi-agent collaboration improves predictive maintenance performance in smart manufacturing environments.

5.1. Dataset and tools

5.1.1. Dataset

The dataset used for this study is publicly available on Zenodo and consists of real-world measurements from a hydraulic system. It includes sensor data collected at different frequencies (1 Hz, 10 Hz and 100 Hz) over 2205 cycles, each lasting 60 s. The dataset contains various sensor measurements related to cooler conditions, valve conditions, internal pump leakage, hydraulic accumulator and system stability. Table 2 summarises the dataset details.[1-13]

[1] https://zenodo.org/records/1323611.

5.1.2. Tools

a. Agentmet4fof Framework

We employ the agent-based framework agentMET4FOF, available on GitHub,[2] specifically designed for manufacturing applications. This open-source framework provides a robust platform for integrating MAS with ML models to facilitate real-time condition monitoring and decision-making in smart manufacturing environments [52]. Its applicability in metrology and intelligent sensor networks has been demonstrated in our previous studies [53–56] and here, we extend its use to PdM for the first time. Key features of agentMET4FOF include:

- Modular Architecture: The framework supports a modular architecture, enabling seamless integration and scalability of various agents. Each agent can perform specialized tasks such as data collection, aggregation, prediction and decision-making.
- Inter-Agent Communication: It provides advanced communication protocols for efficient agent interaction and data exchange. This ensures a cohesive operation where agents can work collaboratively to achieve the system's objectives.
- Real-Time Processing: The framework is optimized for real-time data processing and analysis, making it suitable for monitoring and responding to dynamic changes in manufacturing processes.
- Ease of Integration: agentMET4FOF is designed to integrate easily with existing manufacturing systems and technologies. It supports various data input formats and can interface with multiple sensor types and data sources.

b. Machine learning models

For the ML component, we use BNNs for their capability to handle high-dimensional data and quantify prediction uncertainties. Key tools and libraries utilized in this study include TensorFlow,[3] Keras[4] and PyMC3.[5]

c. Data preprocessing and feature extraction tools

Efficient data preprocessing and feature extraction are crucial for building accurate ML models. Tools and libraries used for these tasks include Pandas,[6] NumPy,[7] SciPy[8] and Scikit-learn.[9]

d. Visualization tools

To visualize the results and monitor the performance of the system, various visualization libraries are used Matplotlib[10] and Seaborn.[11]

e. GANs for data augmentation

To enhance the dataset and improve the robustness of our ML models, we employ GANs for data augmentation. To ensure the quality of the generated data, we validated the GAN outputs using both statistical analysis and visualization techniques. Key distributional metrics (mean, variance and skewness) were compared between real and synthetic datasets. Additionally, we used t-SNE to visualize data clusters and confirm that the synthetic data aligned closely with the structure of real samples. GANs generate synthetic data that mimics the statistical properties of real sensor data [57,58]. This approach helps expand the

[2] https://github.com/Met4FoF/agentMET4FOF.
[3] https://www.tensorflow.org/.
[4] https://keras.io/.
[5] https://pypi.org/project/pymc3/.
[6] https://pandas.pydata.org/.
[7] https://numpy.org/.
[8] https://scipy.org/.
[9] https://scikit-learn.org/stable/.
[10] https://matplotlib.org/.
[11] https://seaborn.pydata.org/.

Table 3
Agents' roles and responsibilities in the MAS framework for PdM.

Agent	Role	Explanation
Sensor agents (DPA)	- Interface with sensors to collect real-time data. - Aggregates and preprocesses diverse data sets, such as historical performance figures, sensor readings and maintenance records, preparing a refined data foundation for subsequent processes.	Operate independently to collect and preprocess data, contributing to the decentralised system as envisaged by DAI. Gather and prepare essential information for the collaborative framework.
Aggregator agent (DPA)	- Aggregates data from Sensor Agents and prepares it for analysis.	Ensures collected data is combined and structured for further processing, providing a comprehensive view for accurate decision-making.
Predictor agent (MTA)	- Runs the trained BNN model on the aggregated data to predict equipment conditions. - It uses pre-processed data to create model's adept at recognising failure-predictive patterns and dynamically adapting to emergent data trends.	Employs RL to continuously improve the predictive model based on new data, ensuring accuracy and responsiveness to changing conditions.
MTA	- Receive the refined dataset from DPAs. - Use GANs to create synthetic data to augment the training datasets. - Train BNNs and other ML models using both real and synthetic data. - Adapt and improve the models based on new data trends and feedback from DMAs.	By incorporating GAN-based data augmentation, the MTAs ensure that the models are trained on a more comprehensive and balanced dataset, improving predictive accuracy and robustness.
DMA	- Uses the predictions and their associated uncertainties to make maintenance decisions. - Orchestrates its counterparts' collaborative efforts. - Facilitates real-time communication, shares vital insights and guides the decision-making process regarding maintenance interventions.	Leverages decentralised data and collaborative efforts to make informed maintenance decisions, integrating predictive insights for system optimisation.
User interface agent	- Provides a visualisation interface for monitoring system performance and making adjustments.	Facilitates the final collaborative process step by presenting data and predictions for stakeholders, enhancing decision-making and operational adjustments.

training dataset, allowing our models to generalize better and perform accurately even with limited original data. Essential tools and libraries for GAN implementation include PyTorch[12] and DCGAN.[13]

5.1.3. Experimental design

a. Data Pre-processing and feature extraction

Before resampling, missing values were imputed using linear interpolation based on neighbouring sensor readings. Additionally, to reduce the impact of sensor noise, a rolling median smoothing filter was applied to each signal. These steps ensured data consistency and reduced the influence of anomalies on model performance. We begin by resampling all sensor data to a consistent frequency of 1 Hz. We extract relevant features from both time and frequency domains for each cycle. These features include statistical measures such as mean, standard deviation, minimum, maximum, sum, median, skewness and kurtosis, resulting in 272 feature inputs per cycle. The extracted features are then normalized for further analysis.

b. ML model training

We use BNNs for multi-task classification to handle the high-dimensional data and quantify the uncertainty of the predictions. The BNN consists of three hidden layers with a node structure of 272–544–272. We train the model using the Bayes by Backprop method for 300 epochs with a learning rate of 0.005, employing the Adam optimizer. The dataset is split into training and testing sets using 5-fold cross-validation to ensure robust performance evaluation. Fig. 4 illustrates the training and validation loss over epochs.

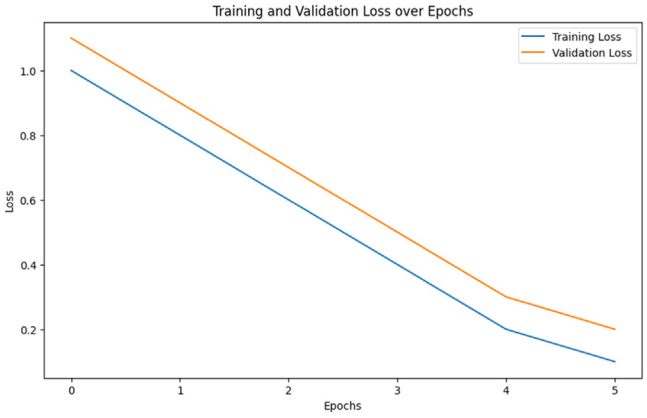


Fig. 4. Training and validation loss over epochs.

c. MAS design

In this study, we propose an MAS framework designed to enhance PdM in smart manufacturing environments. Each agent within the MAS is assigned a specific role, leveraging advanced theoretical concepts such as DAI, RL and collaborative optimisation, which are discussed in Section 3. Table 3 below provides a detailed overview of each agent's roles and tasks. This integration ensures that the MAS operates cohesively, processing real-time data, making accurate predictions and facilitating efficient decision-making for maintenance interventions.

The interaction among these agents is crucial, facilitating a unified approach to data analysis and interpretation, which is expected to enhance PdM accuracy significantly. Sensor data is continuously collected and processed in real-time, with predictions made and decisions adjusted based on current conditions and model uncertainties. Fig. 5 illustrates the flow of data within the proposed framework.

d. Evaluation and Results

The MAS's performance is evaluated based on the accuracy and reliability of the predictions. For instance, the BNN achieved an F1-score of 0.99 for cooler condition classification, with the likelihood of a prediction being accurate given it is certain at 99.70 %. Fig. 6 shows the

[12] https://pytorch.org/.
[13] https://www.tensorflow.org/tutorials/generative/dcgan.

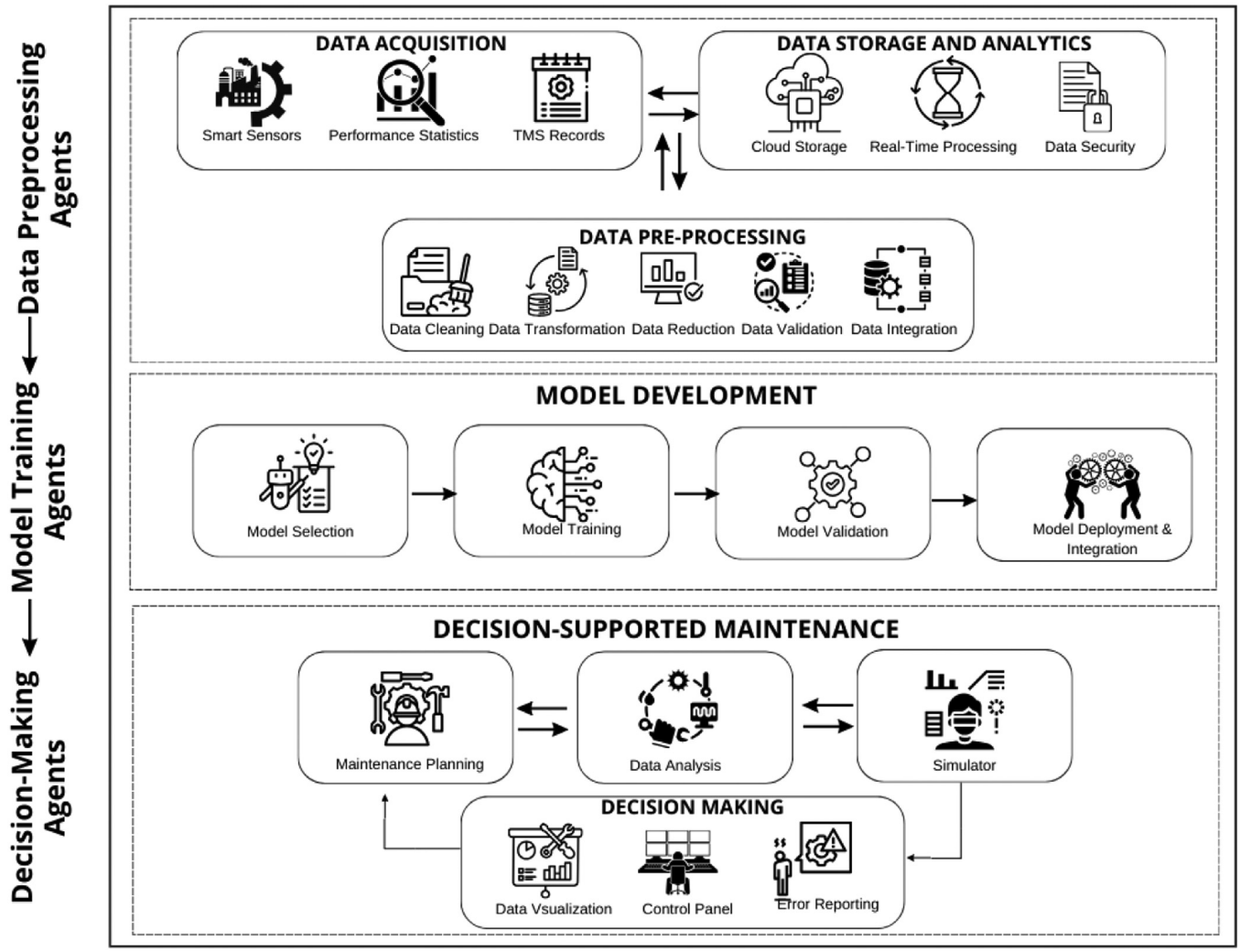


Fig. 5. A visual representation of the proposed framework.

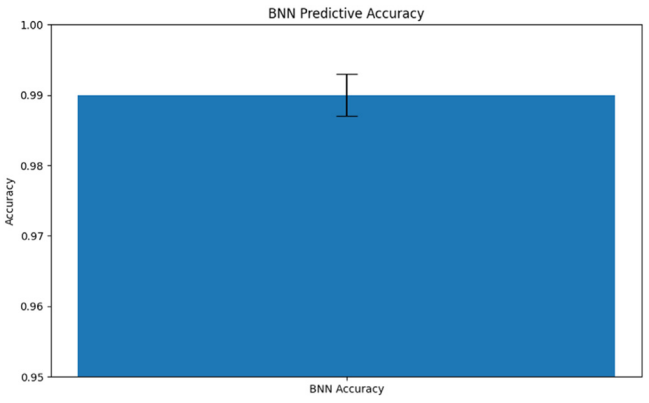


Fig. 6. BNN model's predictive accuracy.

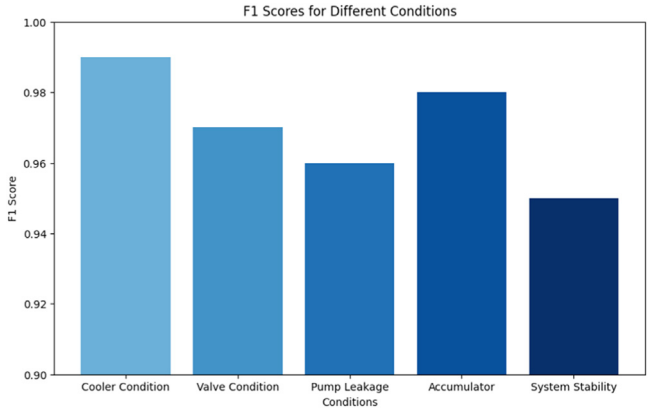


Fig. 7. F1-scores for different conditions.

BNN's predictive accuracy along with the confidence interval. This high level of predictive accuracy, combined with the system's ability to handle uncertainty, demonstrates the effectiveness of the MAS in enhancing PdM practices.

The bar chart in Fig. 7 presents the F1 scores for various conditions monitored in the manufacturing process. The F1 scores for cooler condition, valve condition, pump leakage, accumulator and system stability are 0.99, 0.97, 0.96, 0.98 and 0.95, respectively. These scores highlight the model's effectiveness in accurately predicting different equipment conditions.

To demonstrate the effectiveness of the ML2MAS framework, we benchmarked its performance against several baseline models commonly used for PdM. All models were trained and tested using the same hydraulic dataset.

Table 4
F1-score comparison of the ML2MAS framework and baseline models on the hydraulic dataset.

Model	Architecture	Data strategy	F1-Score (Cooler)	Avg. F1-score (All Classes)
SVM	Supervised ML	Real only	0.85	0.83
Random Forest	Ensemble ML	Real only	0.88	0.86
CNN	DL	Real only	0.91	0.89
Flat RL (DQN)	Single-agent	Real only	0.92	0.90
ML2MAS (Ours)	MAS-RL + BNNs + GANs	Real + Synthetic	0.99	0.97

As shown in Table 4, the ML2MAS framework significantly outperforms traditional ML models, DL approaches and single-agent RL systems. The integration of GAN-augmented training and BNN-based uncertainty handling within a multi-agent framework contributes to this enhanced performance.

e. Integration and future work

The implementation showcases the practical benefits of integrating ML models within a MAS for PdM in smart manufacturing. Future work will involve scaling the system to handle more complex manufacturing environments, exploring additional ML algorithms, refining the agent interactions to improve system performance and adaptability and leveraging GANs for further data augmentation to enhance the robustness and accuracy of the models.

By leveraging the strengths of MAS and advanced ML models, including GANs for data augmentation, this approach aims to improve PdM capabilities in SMSs significantly, leading to reduced downtime, optimized resource usage and enhanced operational efficiency.

6. Discussion and conclusion

The proposed ML2MAS framework, which integrates MAS-RL with BNNs and GANs, offers a novel and practical approach to enhancing PdM in SMSs. In contrast to traditional PdM systems—often centralized and static—ML2MAS supports distributed, adaptive and uncertainty-aware decision-making in dynamic industrial environments.

A central strength of ML2MAS lies in its ability to address data scarcity, a common challenge in industrial settings. The integration of GANs for synthetic data generation mitigates the limitations of limited labelled data, enabling RL agents to generalize across diverse operational scenarios. This improves model robustness, particularly when handling imbalanced or sparse fault classes.

The decentralized, multi-agent architecture further enhances scalability and real-time responsiveness. By assigning specialized roles—such as data sensing, aggregation, prediction, training and decision-making—to autonomous agents, the system supports parallel and cooperative learning. This modular design reduces processing bottlenecks and allows the system to adapt to fluctuating machine behaviours and production conditions.

The case study on a real-world hydraulic dataset demonstrated the practical effectiveness of ML2MAS. The BNN-enhanced model achieved F1-score of 0.99 for cooler condition classification and an average of 0.97 across all fault types. These results validate the system's capacity to accurately detect early signs of degradation and proactively guide maintenance decisions, reducing unplanned downtime and improving operational efficiency.

Compared to conventional PdM approaches, ML2MAS provides several advantages: (1) RL enables continuous adaptation, (2) the MAS structure supports scalable and distributed deployment, (3) GANs enhance predictive accuracy in data-scarce environments and (4) the system delivers near real-time decision-making—essential for minimizing disruptions in SMSs.

In summary, ML2MAS represents a significant advancement in PdM by unifying distributed AI, adaptive learning and synthetic data generation into a cohesive and intelligent maintenance system. Its strong performance in real-world evaluation suggests readiness for deployment across various industrial domains—such as automotive, aerospace and energy—where real-time fault prediction and adaptability are critical for operational success.

7. Limitations and future research

While the ML2MAS framework demonstrates strong performance in PdM through its integration of MASs, RL, GANs and BNNs, several limitations remain.

1. Scalability in complex environments:

ML2MAS has been evaluated on a single use case involving hydraulic systems. Scaling the framework to more complex and larger manufacturing environments—such as those with multiple interconnected production lines—remains an open challenge. Future work should focus on enhancing system scalability and agent coordination strategies in large-scale industrial applications.

2. Algorithm integration and extension:

The current implementation focuses on PPO for RL and DCGANs for data augmentation. Future research could explore the integration of additional ML, RL and GAN variants to improve adaptability, learning speed and model robustness under different operational conditions.

3. Agent communication and coordination:

Refining inter-agent communication remains an important area for future improvement. As the number of agents increases, maintaining efficient and reliable collaboration becomes more complex. Research into lightweight protocols and decentralized coordination mechanisms can help optimize performance while minimizing latency and overhead.

4. Interpretability of decisions:

Although BNNs provide uncertainty estimates, the decision-making process of agents could benefit from greater transparency. Incorporating explainable AI (XAI) techniques may help operators understand and trust the system's autonomous decisions, especially in high-risk environments [59].

5. Synthetic data validation:

While GANs help overcome data scarcity, further evaluation is needed to ensure the realism and reliability of synthetic data. Future work should explore statistical testing, adversarial validation and domain expert assessments to ensure synthetic data quality does not compromise predictive performance.

6. Integration of emerging technologies:

Additionally, integrating emerging technologies such as blockchain for secure data sharing or digital twins for real-time system simulation could further enhance the framework. Blockchain can provide a secure and transparent method for sharing sensor data across different manufacturing sites, improving collaboration while ensuring data integrity

[60]. Digital twins, on the other hand, would allow for real-time manufacturing process simulations, enabling more accurate PdM by testing various "what-if" scenarios before applying changes to the actual system [61].

7. Cross-domain applicability:

Lastly, while this paper focuses on a single-use case (hydraulic systems), the proposed framework is highly adaptable. It can be extended to other industries, such as automotive, aerospace and energy. Future research may evaluate the scalability and performance of ML2MAS in these domains, especially when handling larger datasets and more intricate system interdependencies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Shadia Yahya Baroud: Writing – original draft, Resources, Formal analysis, Writing – review & editing, Software, Investigation, Conceptualization, Visualization, Methodology, Data curation. **Nor Adnan Yahaya:** Validation, Investigation, Methodology, Supervision, Conceptualization.

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